α/D_0: The Universal CRF Coupling

One Dimensionless Ratio Beneath Gravity, Cosmology, and Quantum Action

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Synthesises CRF Gravitational Constant, CRF KCoupling, CRF GtRate. No new axioms. Pattern identification only.

Abstract. A single dimensionless ratio

$$\mathcal{G} \equiv \frac{\alpha}{D_0} = \frac{\ln 2}{n(1 - e^{-1/n})} \approx 0.7373$$

appears as the controlling coefficient in every gravitational, cosmological, and quantum-action expression in CRF. It is a *pure function of* $n = 8$ (the SO(3) symmetry class), with no other inputs.

This document identifies all occurrences of $\mathcal{G}$ across the CRF derivation chain, derives $\mathcal{G}$ from n alone via Identity K2, and shows that five apparently different physical quantities are in fact five expressions of the same informational ratio: the fraction of vacuum information capacity consumed per δ-operation.

New consequence: the Dirac large-numbers coincidence $G \cdot M_{\text{proton}}^2/\hbar c \sim 10^{-40}$ is explained as $\mathcal{G}^2$ evaluated at cosmological N_{eff}, without anthropic reasoning.

The Discovery: One Ratio, Five Roles

Discovery

Reviewing CRF Gravitational Constant, CRF KCoupling, and CRF GtRate, the ratio $\mathcal{G} = \alpha/D_0$ appears as the controlling factor in five distinct physical expressions, each derived independently:

Physical role	CRF expression	Source
C5 coupling k	$k = \alpha^2/D_0 = \alpha \cdot \mathcal{G}$	CRF Gravitational C
Gravitational constant G	$G = (\alpha \cdot c^2)/D_0 = \mathcal{G} \cdot c^2$	CRF Gravitational C
$T_{\mu\nu}$ coupling $\mathcal{K}$	$\mathcal{K} = c^2/(\alpha^2 D_0) = c^2/(\alpha \cdot \mathcal{G}^{-1} \cdot D_0^2) = c^2\mathcal{G}/(\alpha\, D_0^2)$	CRF KCoupling
Cosmic rate $\lvert\dot{G}/G\rvert$	$\lvert\dot{G}/G\rvert = \mathcal{G} \cdot H_0$	CRF GtRate
Dark energy EOS w	$w = -1 + \mathcal{G}/3 \quad \text{(see §4)}$	CRF GtRate

Interpretation: $\mathcal{G}$ is not a coupling constant in the conventional sense. It is the ratio of the vacuum baseline constraint $\alpha = \ln 2$ (the information content of one δ-operation) to the total information scale of the vacuum D_0 (how large $\mathcal{P}$ is in KL-divergence units). It measures what fraction of the vacuum information capacity is consumed by a single δ-event.

$\mathcal{G}$ as a Pure Function of n

From Identity K2 (closed, 0.03% gap):

$$D_0 = n\left(1 - e^{-1/n}\right)$$

Combined with $\alpha = \ln 2$ (Session 8, exact):

Theorem 1 ($\mathcal{G}$ from n alone)**.**

$$\mathcal{G} \;=\; \frac{\alpha}{D_0} \;=\; \frac{\ln 2}{n(1 - e^{-1/n})}$$

This is a pure function of $n = 8$ (the SO(3) symmetry class, CRF SO3 Proof). No other inputs.

Numerically:

$$\mathcal{G} = \frac{0.693147}{8 \times (1 - e^{-1/8})} = \frac{0.693147}{8 \times 0.117503} = \frac{0.693147}{0.940025} = 0.73737\ldots$$

Corollary 1 (Connection to K^*)**.** *From K1 ($K^* \approx D_0/n$, 0.43% gap):*

$$\mathcal{G} = \frac{\alpha}{D_0} = \frac{\alpha}{nK^*} = \frac{\ln 2}{nK^*}$$

This is the form used in CRF GtRate for the cosmic variation rate. Both expressions are the same $\mathcal{G}$ to 0.43%.

Expansion in $1/n$

For large n, $1 - e^{-1/n} \approx 1/n - 1/(2n^2) + \ldots$, so:

$$D_0 \approx 1 - \frac{1}{2n} + O(1/n^2) \qquad \Rightarrow \qquad \mathcal{G} \approx \ln 2 \cdot \left(1 + \frac{1}{2n} + O(1/n^2)\right)$$

At $n = 8$: $\mathcal{G} \approx \ln 2 \cdot (1 + 1/16) = 0.6931 \times 1.0625 = 0.7364$ (vs exact 0.7374, 0.1% gap). The leading term is simply $\ln 2 \approx 0.693$; $\mathcal{G}$ is $\ln 2$ corrected upward by $1/(2n) = 6.25\%$ for the finite mode count.

Physical Interpretation: Fraction of Vacuum Capacity

$\mathcal{G}$ has a direct informational interpretation:

Physical meaning of $\mathcal{G}$

Numerator $\alpha = \ln 2$: The information content of one δ-operation — the minimum KL divergence introduced by a single resolution event. This is the "cost" of one distinction.

Denominator $D_0 = n(1-e^{-1/n})$: The total information capacity of the vacuum in one mode-cycle — how much KL divergence the system can accumulate across n modes before the resolution threshold is reached.

$$\mathcal{G} = \frac{\text{cost of one } \delta\text{-operation}}{\text{total vacuum capacity per cycle}}$$

This ratio determines how strongly each δ-event "dents" the vacuum. A large $\mathcal{G}$ means each event consumes a large fraction of capacity $\rightarrow$ strong gravity, fast cosmic evolution, large quantum action. For our universe with $n = 8$: $\mathcal{G} \approx 0.74$ — each R-event consumes 74% of one mode's information capacity.

The Five Roles Unified

Role 1 — C5 coupling: $k = \alpha \cdot \mathcal{G}$

The C5 ODE coefficient $k = \alpha^2/D_0 = \alpha \cdot \mathcal{G}$ sets the rate at which the constraint field $\mathcal{C}$ saturates per unit radial distance at the information horizon. $\mathcal{G}$ enters because each R-event advances $\mathcal{C}$ by α (one δ-bit), but the vacuum capacity D_0 determines how far that advance propagates spatially.

Role 2 — Gravitational constant: $G = \mathcal{G} \cdot c^2$

$$G = \frac{\alpha \cdot c^2}{D_0} = \mathcal{G} \cdot c^2$$

Gravity is weak because $\mathcal{G} < 1$ and c^2 must be converted to SI units via the Planck length. In natural units where $c = 1$: $G = \mathcal{G}$, making $\mathcal{G}$ the literal gravitational coupling in CRF's information geometry.

Role 3 — $T_{\mu\nu}$ coupling: $\mathcal{K}$

$$\mathcal{K} = \frac{c^2}{\alpha^2 D_0} = \frac{\mathcal{G}}{\alpha^2 D_0^2} \cdot \alpha^2 \cdot \frac{c^2}{\mathcal{G}} = \frac{c^2}{\alpha\, D_0 \cdot \mathcal{G}^{-1} \cdot D_0}$$

More cleanly: $\mathcal{K} = G/(\alpha^2) = \mathcal{G} \cdot c^2/\alpha^2$. The $T_{\mu\nu}$ coupling is G divided by the square of the vacuum baseline — another expression of $\mathcal{G}$.

Role 4 — Cosmic rate: $|\dot{G}/G| = \mathcal{G} \cdot H_0$

From CRF GtRate:

$$\left|\frac{\dot{G}}{G}\right| = \frac{\ln 2}{nK^*}\, H_0 = \mathcal{G} \cdot H_0$$

The fractional rate of cosmic G-variation equals the universal coupling times the Hubble rate. Since $\mathcal{G} \approx 0.74$, the relaxation timescale is $1/(\mathcal{G} H_0) \approx 1.36\, t_H$ — slightly longer than the Hubble time.

Role 5 — Dark energy equation of state

From $\rho_\Lambda \propto e^{-\mathcal{G}\, H_0\, t}$ and standard FLRW:

$$w = -1 + \frac{\mathcal{G}}{3} = -1 + \frac{\ln 2}{3\, n\, (1 - e^{-1/n})} \approx -1 + 0.246 = -0.754$$

The factor of 3 is the spatial dimension $d_s = 3$ from SO(3). So more precisely:

$$\boxed{w = -1 + \frac{\mathcal{G}}{d_s} = -1 + \frac{\alpha}{d_s\, D_0}}$$

New Consequence: The Dirac Coincidence from $\mathcal{G}$

The coincidence

Dirac noted that $GM_p^2/(\hbar c) \approx 10^{-40}$, while the ratio of the Hubble time to the atomic unit time is also $\sim 10^{40}$. This "large numbers coincidence" has been unexplained for 90 years.

CRF derivation

In CRF, $G = \mathcal{G} \cdot c^2$ (natural units) and $\hbar_{\text{CRF}} = \varepsilon_0 \cdot D_0/\ln 2 = \varepsilon_0/\mathcal{G}$ (Track A). The proton mass in CRF is $M_p \sim m_{\text{fire}} \cdot \kappa/c^2$ where $m_{\text{fire}} = \varphi$ (K6, closed).

The ratio:

$$\frac{GM_p^2}{\hbar c} = \frac{\mathcal{G} \cdot c^2 \cdot M_p^2}{(\varepsilon_0/\mathcal{G}) \cdot c} = \frac{\mathcal{G}^2 \cdot c \cdot M_p^2}{\varepsilon_0}$$

This is proportional to $\mathcal{G}^2/\varepsilon_0$. With $\mathcal{G} \approx 0.74$ and $\varepsilon_0 = 0.00755$: $\mathcal{G}^2/\varepsilon_0 \approx 0.547/0.00755 \approx 72.5$.

The large number 10^{40} arises from c and M_p in SI units, but the *dimensionless structure* is:

$$\frac{GM_p^2}{\hbar c} \propto \frac{\mathcal{G}^2}{\varepsilon_0} \approx 72.5 \qquad \text{(in CRF natural units)}$$

The Dirac coincidence — that this ratio equals the number of atomic time units in t_H — is then:

$$\frac{t_H}{t_{\text{atomic}}} \sim \frac{1}{\mathcal{G}H_0 \cdot t_{\text{atomic}}} \sim \frac{1}{\mathcal{G}^2/\varepsilon_0}$$

in CRF natural units: **both ratios are $\mathcal{G}^2/\varepsilon_0$**. The coincidence is not coincidental — it is the statement that G, $\hbar$, and H_0 are all determined by $(\mathcal{G}, \varepsilon_0, n)$, and these three quantities are fixed by δ.

Complete Occurrence Table

Quantity	Expression in $\mathcal{G}$	Numerical value
$\mathcal{G}$ itself	$\ln 2\ /\ [n(1-e^{-1/n})]$	0.7374
k (C5 coupling)	$\alpha \cdot \mathcal{G}$	0.5113
G (natural units)	$\mathcal{G} \cdot c^2$	$0.7374\, c^2$
$\mathcal{K}$ ($T_{\mu\nu}$ coupling)	$\mathcal{G}\, c^2/\alpha^2$	$1.534\, c^2$
$\lvert\dot{G}/G\rvert$	$\mathcal{G} \cdot H_0$	$0.737\, H_0$
$w_{\mathrm{DE}} + 1$	$\mathcal{G}/d_s$	0.246
$\mathcal{G}^2/\varepsilon_0$ (Dirac ratio)	$\alpha^2/(D_0^2\, \varepsilon_0)$	72.5 (CRF units)

$\mathcal{G}$ in the δ-Chain Structure

$$
\begin{array}{rcl}
\delta \text{ binary, SO(3)} & \xrightarrow{n=8} & D_0 = n(1-e^{-1/n}), \quad \alpha = \ln 2 \\
\alpha,\ D_0 & \longrightarrow & \boxed{\mathcal{G} = \alpha/D_0 = 0.7374} \\
\mathcal{G} & \xrightarrow{\times c^2} & G = \mathcal{G}c^2 \quad \text{(gravity)} \\
\mathcal{G} & \xrightarrow{\times H_0} & |\dot{G}/G| = \mathcal{G}H_0 \quad \text{(cosmic rate)} \\
\mathcal{G} & \xrightarrow{/d_s} & w + 1 = \mathcal{G}/d_s \quad \text{(dark energy)} \\
\mathcal{G} & \xrightarrow{2/\varepsilon_0} & \mathcal{G}^2/\varepsilon_0 \quad \text{(Dirac ratio)} \\
\text{All} & \longrightarrow & \textbf{One function of } n \textbf{ alone}
\end{array}
$$

Predictions Unified Under $\mathcal{G}$

All CRF cosmological predictions share coefficient $\mathcal{G} = 0.737$

P-cosm-1. $|\dot{G}/G| = 0.737\, H_0 \approx 5.1 \times 10^{-11}\ \mathrm{yr}^{-1}$ *(GW sirens, PTA)*

P-cosm-2. $|\dot{\Lambda}/\Lambda| = 0.737\, H_0$ *(DESI DR2, Euclid)*

P-cosm-3. $\dot{G}/G = \dot{\Lambda}/\Lambda$ exactly (ratio = 1.000) *(cross-correlation)*

P-cosm-4. $w_{\mathrm{DE}} = -1 + 0.737/3 = -0.754$ *(DESI, Euclid)*

P-cosm-5. $G(z)/G_0 = [1+z]^{0.737}$ for matter-domination *(multi-epoch GW)*
If any measurement gives a coefficient significantly different from 0.737, the CRF mechanism for P-growth is falsified — without affecting the algebraic structure (which derives $\mathcal{G}$ exactly from n).

Summary

The ratio $\mathcal{G} = \alpha/D_0 = \ln 2\,/\,[n(1-e^{-1/n})] = 0.7374$ is the universal CRF coupling constant. It is a pure function of $n = 8$ (the SO(3) mode count, derived from Axiom 0 and Axiom 11).
It controls:

- The C5 field equation ($k = \alpha\mathcal{G}$)
- Newton's constant ($G = \mathcal{G}c^2$ in natural units)
- The $T_{\mu\nu}$ coupling ($\mathcal{K} = \mathcal{G}c^2/\alpha^2$)
- The cosmic rate of G and Λ variation ($\mathcal{G}H_0$)
- The dark energy equation of state ($w = -1 + \mathcal{G}/d_s$)
- The Dirac large-numbers ratio ($\mathcal{G}^2/\varepsilon_0$)

Physical meaning: $\mathcal{G}$ is the fraction of the vacuum's information capacity consumed by a single δ-operation. It is 74% because the universe has $n = 8$ modes and the information per event ($\ln 2$) is 74% of the total mode capacity (D_0).
Everything gravitational, cosmological, and action-like in CRF is one number wearing different clothes.